

Turibius Rozario

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Objective

Graduate-level Mechanical Engineer specializing in energy and sustainability, seeking opportunities to support the design, analysis, and implementation of renewable energy systems that reduce environmental impact and foster inclusive, just transitions to a low-carbon future.

Education

Masters (Thesis) in Mechanical Engineering (ME)
Boston University (BU)

Expected January 2027
3.95/4.00 GPA

BS in ME, Minor in Computer Science (CS)
University of Maryland, Baltimore County (UMBC)

May 2025
3.91/4.00 GPA

Audited Courses: Energy Within Environmental Constraints (HarvardX, EdX), Fundamentals of Fluid Power (University of Minnesota, Coursera), Energy Sources for the Future (UMBC)

Research Experience

Predicting Critical Operating Conditions for Solid Oxide Cells

January 2026 – Present

BU, ME Department

Mentor: Dr. Srikanth Gopalan (sgopalan@bu.edu, available to contact)

- Currently developing physics-based models coupling electrochemistry, gas transport, and mechanical stress to predict critical overpotential and current density in solid oxide fuel and electrolysis cells, quantifying how oxygen partial-pressure buildup at the oxygen-electrode/electrolyte interface drives fracture and constrains safe, high-performance operation for green-hydrogen systems.

Integration of Controls and Neural Networks

November 2021 – January 2025

UMBC, ME Department

Mentor: Dr. Ankit Goel (ankgoel@umbc.edu)

- Built and validated end-to-end training and optimization pipelines in MATLAB, TensorFlow, and PyTorch to advance control algorithms for difficult-to-model Physics problems.
- Constructed tiny (32×4) yet accurate neural networks capable of predicting chaotic trajectories by integrating Extended Kalman filters, in contrast to large (several thousands) neural networks.

Magnesium Extraction Methods from Seawater

Summer 2024

University of Wisconsin, Madison (UW), ME Department

Mentor: Dr. Michael Wagner (mjwagner2@wisc.edu, available to contact)

- Conducted a system-level, techno-economic literature review of seawater-based magnesium extraction and co-recovery of critical minerals, assessing precipitation and metal-production routes on energy demand and waste-heat integration to provide a decision-focused framework for environmentally preferable alternatives to open-pit mining.

Design of a Hardware-in-the-Loop Test System for Wave Energy Harvesting

Summer 2023

University of Minnesota, Twin Cities (UMN), ME Department

Mentor: Dr. James Van de Ven (vandeven@umn.edu)

- Developed an analytical scaling framework for a hardware-in-the-loop lab test of a wave-energy system that co-generates electricity and freshwater, using fluid power and energy balance calculations to map full-scale offshore conditions to equivalent laboratory pressures, flows, and loads.

Publications

Oveissi, P., **Rozario, T.**, and Goel, A. “A Novel Neural Filter to Improve Accuracy of Neural Network Models of Dynamic Systems.” 2025 IEEE Conference on Control Technology and Applications (CCTA), 2025, pp. 705–710. <https://doi.org/10.1109/ccta53793.2025.11151364>.

Rozario, T., Oveissi, P., and Goel, A. “Matrix-Based Representations and Gradient-Free Algorithms for Neural Network Training.” 2024 International Conference on Machine Learning and Applications (ICMLA), 2024, pp. 325–332. <https://doi.org/10.1109/icmla61862.2024.00050>.

Presentations

Undergraduate Research and Career Advancement Day , UMBC “UMBC Multidisciplinary Capstone: Design and Development of an Autonomous Fixed-Wing UAV for Long-Range Payload Delivery”	April 16, 2025
Summer Undergraduate Research Experience Poster Session , UW “Magnesium Extraction Methods from Seawater”	July 31, 2024
Undergraduate Research and Career Advancement Day , UMBC “Modelling Dynamic Systems Using Neural Networks”	April 10, 2024
Summer Undergraduate Research Expo , UMN “Design of a Lab-Scale Ocean Wave-Powered Desalination System”	August 10, 2023
Undergraduate Research and Career Advancement Day , UMBC “A Tutorial on Neural Networks and Gradient-free Training”	April 12, 2023

Co-curricular Activities

UMBC American Institute of Aeronautics and Astronautics (AIAA) <i>External Student Advisor</i> <i>Student Unmanned Aerial Systems (SUAS) Captain</i> <i>Design, Build, Fly (DBF) Project Lead</i>	July 2025 – Present May 2024 – June 2025 May 2023 – May 2025
• Currently advise the SUAS captain and DBF lead in organizational management, engineering design and construction decision making, and ensuring workplace safety. • Guided 20~30 members from diverse academic and economic backgrounds without prior aviation experience by lecturing on aerodynamics, training them in carbon fiber manufacturing, teaching flying techniques, and co-creating a shared space for pursuing aviation interests despite no aerospace track/department available at school. • Led both teams to their very first successful flights at the SUAS and DBF international competitions, and placing 17th-place overall finish out of 81 teams in SUAS.	
BU Unmanned Aerial Vehicles (UAV), <i>Graduate Student Advisor</i>	September 2025 – Present
• Currently advising club members on a weekly basis on management, funding, and aircraft building to head to their very first SUAS competition.	
Student Government Association, <i>First Year Ambassador</i>	September 2021 – May 2022
• Hosted campus-wide wellness events for over 200 people • Initiated proposals with university stakeholders for improving campus COVID-resiliency	

Skills

Programming	Python, MATLAB & Simulink, L ^A T _E X, HTML / CSS, Typst, C++, Mathematica
Software	GNU/Linux, COMSOL, PyTorch, Keras, SolidWorks, PHREEQC, Inkscape, ArduPilot, OpenCV, OpenIPV
Hardware	Arduino, Raspberry Pi, BeagleBone, Sensor Modules
Technical abilities	3D Printing, Power Tools, Soldering, Hand Lamination, Model Aircraft Pilot

Awards & Honors

Student Leadership Award , UMBC	May 2025
Sustained Research Award , UMBC	October 2023 – May 2024
President’s List , UMBC	January 2022 – January 2024
S-STEM Scholar , UMBC	June 2022 – June 2023